FNU Prayoga

425-520-3074 | Corvallis, OR | prayoga23.yoga@gmail.com | LinkedIn: linkedin.com/in/prayoga-e-789868138
Portfolio: pprayoga.github.io/Portfolio

Relevant Projects

AI Invention Extraction & Prior-Art Analysis (InventionID) — M.S. Capstone, Oregon State University
Sept. 2025 – Jun. 2026

- As part of a 3-person team, built an agentic LLM pipeline that extracts candidate inventions from technical documents and runs a first-pass patentability check (PDF parsing → summarization → embeddings → patent-query generation → prior-art retrieval), with RAG over a FAISS index for cited evidence.
- Owned the prior-art search and patentability screening (§101 subject-matter eligibility) — the system's novelty-analysis core — querying Google Patents and Google Scholar via the SerpAPI and Semantic Scholar APIs.
- Contributed to shared components, including long-context inference on AWS Bedrock with prompt caching to reduce token cost and latency.

Diary Agent — Independent Project
Jul. – Aug. 2026

- Built and deployed a personal knowledge system that converts daily journal entries into structured meeting recaps and a to-do list via a multi-stage LLM pipeline.
- Exposed the diary and an auto-tagged research-paper library through a Model Context Protocol (MCP) server for natural-language retrieval (SQLite + FTS5 full-text and tag search); works with both ChatGPT and the Claude desktop app.
- Shipped to production on AWS EC2 (FastAPI + Streamlit) behind a Cloudflare Tunnel with Access auth and a cron-driven nightly recap; profiled memory and right-sized the instance (t3.medium → t3.micro), cutting monthly cost ~4×.

Work Experience

Field Service Representative — Solar Turbines Inc.
Jakarta, Indonesia · May 2018 – Jun. 2023

- Commissioned, inspected, and troubleshot gas-turbine packages on customer sites — frequently the sole engineer on site, owning the customer relationship end to end.
- Led cross-functional teams of ~6 technicians through installation, calibration, and start-up under tight field timelines.
- Diagnosed and resolved issues spanning mechanical, electrical, PLC, instrumentation, and process systems to keep customer equipment running safely and to spec.
- Authored technical reports and communicated findings and best practices to customers and teams, translating deep engineering detail for non-technical stakeholders.

Research

Graduate Research — Data-Centric ML / Minimal Imputation — Oregon State University
Advisor: Arash Termehchy · Aug. 2024 – Present

- Developed ML-based minimal-imputation strategies that reduce how much missing data must be imputed while maintaining or improving downstream classification accuracy.
- Benchmarked imputation methods (KNN, MICE, diffusion-based TabCSDI, LLM baselines) in PyTorch with automated metric tracking; ran experiments on HPC (SLURM).
- Work published at VLDB 2026 (demonstration), arXiv, and a NeurIPS 2025 workshop (see Publications).

Publications

1. Cheng Zhen, **Prayoga**, Nischal Aryal, Arash Termehchy, Alireza Aghasi. “Minimal Data Cleaning for Model Training by MinPrep.” VLDB 2026 (Demonstration).
2. Cheng Zhen, Nischal Aryal, Arash Termehchy, **Prayoga**, Garrett Biwer, Sankalp Patil. “Learning Accurate Models on Incomplete Data with Minimal Imputation.” arXiv:2503.13921, 2025.
3. Cheng Zhen, Nischal Aryal, Arash Termehchy, **Prayoga**, Garrett Biwer. “Minimal Repairs for Learning Over Incomplete Data.” NeurIPS 2025 Workshop: Reliable ML from Unreliable Data (Poster).

Other Projects

- **Language Model Data Extraction —** Reproduced Carlini et al. (2021); fine-tuned GPT-2 on synthetic PII and quantified leakage via perplexity-based extraction (PyTorch, SLURM).
- **Data Poisoning via Influence Functions —** Targeted class-poisoning on CIFAR-10 using TRACIn influence scoring with FGSM/PGD on ResNet-56 (PyTorch).
- **Robustness Evaluation with AutoAttack —** Benchmarked classifiers on CIFAR-10 under AutoAttack (PGD, CE-PGD, FAB, Square); compared adversarially trained vs standard models.

Skills

ML & AI: LLM agents, RAG, embeddings, prompt & context engineering, fine-tuning, adversarial robustness

Frameworks & Tools: PyTorch, scikit-learn, pandas, NumPy, FAISS, AWS Bedrock, Git, HPC / SLURM

Languages: Python (primary), Bash, SQL

Field Engineering: Gas-turbine commissioning, PLC, electrical, instrumentation, process, team leadership, customer-facing delivery

Education & Certification

Oregon State University, Corvallis, OR
Sept. 2023 – Jun. 2026
Master of Science in Artificial Intelligence, GPA: 3.87

Texas A&M University – Corpus Christi, Corpus Christi, TX
Jan. 2014 – Dec. 2017
Bachelor of Science in Mechanical Engineering, GPA: 3.89, Dean's List

Certification: Fundamentals of Engineering (FE)